Université Internationale de Rabat

School of Computer Science and Digital Engineering - Big Data & AI

Project Report

Barcode and QR Code Scanner and Localizer

Design of a Computer Vision System for Automatic Detection and Decoding

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1 Introduction

QR codes and barcodes are widely used for storing and retrieving information in domains such as retail, logistics, and access control. They provide a fast and efficient way to encode data that can be easily scanned and processed.

The objective of this project is to develop a computer vision system capable of detecting and decoding both QR codes and barcodes from images. The detection pipeline is implemented from scratch using classical image processing techniques.

The system is divided into two main tasks: detection and decoding. Barcode detection relies on gradient analysis to highlight vertical patterns, while QR code detection is based on identifying square-shaped structures and characteristic finder patterns. Once detected, regions are preprocessed and decoded using appropriate tools: `pyzbar` for barcodes and OpenCV's QRCode detector for QR codes.

The results demonstrate the effectiveness of traditional computer vision methods, while highlighting challenges such as noise, image quality, and multiple detections.

2 Methodology

2.1 Barcode Detection

The barcode detection process is based on gradient analysis in order to highlight the vertical line patterns that characterize barcodes.

First, the input image is converted to grayscale to simplify the processing. Then, a **Gaussian blur** is applied to reduce noise. Next, **Sobel operators** are used to compute the gradients in both horizontal and vertical directions. The difference between these gradients ($|G_x| - |G_y|$) is used to emphasize vertical structures.

The resulting image is then thresholded to obtain a binary representation. A morphological closing operation is applied to connect broken parts of the barcode region. Finally, connected component analysis is performed to extract candidate regions, and bounding boxes are generated based on geometric properties.

2.2 Barcode Decoding

Once barcode regions are detected, they are cropped from the original image. To improve decoding performance, preprocessing steps such as resizing, contrast enhancement, and thresholding are applied.

The decoding is then performed using the `pyzbar` library, which extracts the encoded information from the barcode.

2.3 QR Code Detection

The QR code detection process focuses on identifying the characteristic square finder patterns.

The image is first converted to grayscale and smoothed using Gaussian blur. Then, adaptive thresholding is applied to handle variations in lighting conditions. Connected component analysis is used to extract candidate regions.

These regions are filtered based on geometric constraints such as aspect ratio and fill ratio to retain square-like shapes. Triplets of such squares are then analyzed to detect valid QR code finder patterns. A final bounding box is computed from the detected triplet.

2.4 QR Code Decoding

After detecting QR code regions, they are cropped and optionally preprocessed. Decoding is performed using OpenCV's QRCode detector, which returns the encoded information if a valid QR code is present.

3 Results

This section presents the results obtained using the proposed system for detecting and decoding barcodes and QR codes, as well as the final integrated application.

3.1 Barcode Detection

The barcode detection algorithm successfully identifies regions containing barcodes using gradient-based analysis. The detected region is highlighted using a bounding box.

Figure 1: Barcode detection using gradient-based method

3.2 Barcode Decoding

After detection, the barcode region is cropped and decoded using the pyzbar library. The system correctly extracts the encoded numerical information.

Figure 2: Barcode decoding result showing extracted EAN13 code

3.3 QR Code Detection

The QR code detection method successfully locates QR codes by identifying square finder patterns. The detected QR code is enclosed within a bounding box.

Figure 3: QR code detection using geometric analysis

3.4 QR Code Decoding

The detected QR code is decoded using OpenCV's QRCode detector. The system successfully retrieves the embedded URL.

Figure 4: QR code decoding result showing extracted link

3.5 Integrated Application (Barcode + QR Scanner)

A graphical user interface was developed to integrate both barcode and QR code detection and decoding into a single application.

Figure 5: User interface of the unified barcode and QR scanner

The application allows users to upload an image and perform detection and decoding in a single step. The results are displayed with bounding boxes and decoded information.

```
=== BARCODE RESULTS ===

Barcode 1
Box     : (30, 340, 177, 788)
Decoded : True
Text    : 5123451234567
Type    : EAN13
```

```
=== QR RESULTS ===

QR 1
Box     : (0, 0, 240, 268)
Decoded : True
Text    : https://qrco.de/signup-code
```

Figure 6: Final result showing both QR code and barcode detection and decoding

4 Challenges

During the development of this project, several challenges were encountered.

One of the main difficulties was related to barcode decoding using the `pyzbar` library, particularly due to dependency issues and environment configuration. Ensuring proper installation and compatibility required additional effort.

Another challenge was the occurrence of multiple detections and false positives, especially when processing noisy or complex images. Fine-tuning parameters such as threshold values and morphological operations was necessary to improve detection accuracy.

The integration of both barcode and QR code pipelines into a unified system also presented difficulties. Combining two different detection and decoding approaches while maintaining consistency in the output required careful design.

Finally, the performance of the system was sensitive to image quality, including lighting conditions, resolution, and noise, which affected both detection and decoding accuracy.

5 Conclusion

In this project, a complete computer vision system for detecting and decoding barcodes and QR codes was successfully developed. The detection process was implemented from scratch using classical image processing techniques, while decoding was performed using specialized tools.

The results demonstrate that the proposed approach is effective in identifying and decoding multiple types of codes within a single image. Additionally, the development of a unified application highlights the practical usability of the system.

However, some limitations remain, particularly regarding robustness under challenging conditions. Future improvements could include the integration of deep learning-based methods to enhance detection accuracy and reliability.